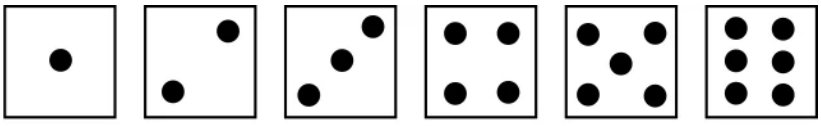
Discrete random variables

- Let X be a random variable. If it is discrete then it may be finite or countably infinite i.e. possible values maybe $x_{1,}\, x_{2,}\, x_{3,}$ ... and so on.
- **For instance,** consider the example of tracking the number of jobs arriving weekly at a job shop. Here, the random variable X represents the number of jobs arriving each week, with possible values ranging from 0 onward, denoted by $R_x = \{1,\ 2, 3..\}$
- When dealing with discrete random variables like X each possible outcome x_i in the range space R_x has an associated probability $p(x_i) = P(X = x_i)$. These probabilities must adhere to two fundamental conditions:

  1. $p(x_i) > 0\ for\ all\ possible\ i$
  2. The sum of all possible values of X is 1 $\sum_{i=1}^{\infty} p(x_i) = 1$

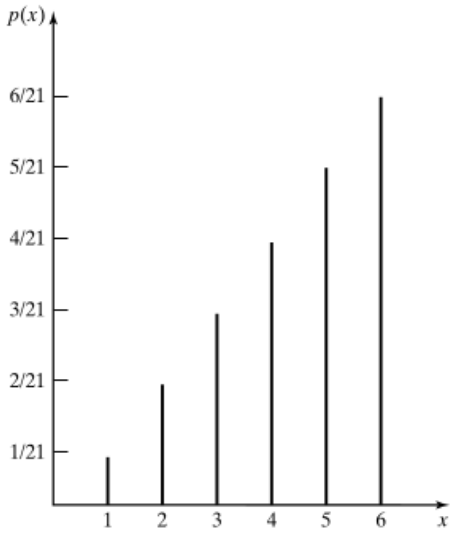
- The set of all pairs of $(p(x_i),\ x_i)$ where *i = 1, 2, 3...* Constitutes the probability distribution of X. $p(x_i)$ here is called the probability mass function (pmf)
- **Another example**, Suppose we're tossing a single die, and we define the random variable X as the number of spots on the top face of the die after it's tossed. In this case, the possible outcomes for denoted as R_x, are {1, 2, 3, 4, 5, 6} since these are the numbers of spots on a standard six-sided die.



- Now, let's assume that the die is loaded, meaning that the probability of a specific face landing up is proportional to the number of spots showing on that face. The discrete probability distribution for this experiment is as follows:

x_i	P
1	1/21
2	2/21
3	3/21
4	4/21
5	5/21
6	6/21

- These probabilities satisfy the conditions we mentioned earlier
- The probability distribution is illustrated below



Continuous random variables

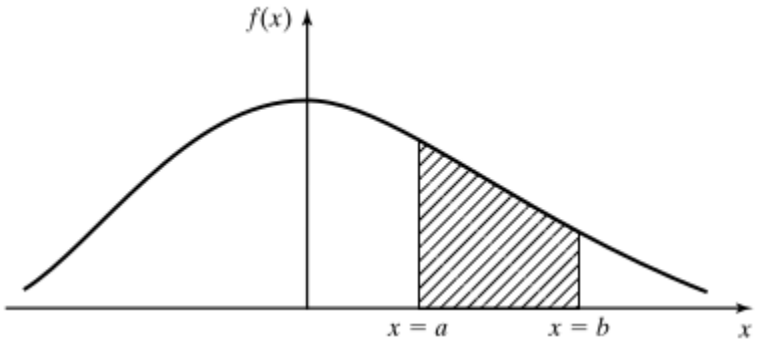
- If the range of possible values of a random variable is an interval or a collection of intervals, then X is called a continuous random variable. For continuous random variable the probability that lies within the interval [a, b] is given by:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

- $f(x)$ is the probability density function (pdf) and it satisfies the following conditions:

1. $f(x) > 0$ for all x in the range of X
2. The integral of $f(x)$ over the range of X is 1
3. $f(x) = 0$ if x is not in the range of X.

- For a continuous random variable X, the probability of it taking on any specific value say, x_0 is always $P(X = x_0) = 0$. This implies that the probability of observing any exact value is infinitesimally small.



The shaded area represents probability that X lies in [a, b]

- For instance: the lifetime of a device used to inspect cracks in aircraft wings is represented by the continuous random variable X, which takes on all values in the

range $x > 0$. The probability density function (pdf) of the device's lifetime, denoted by $f(x)$ is given by:

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2} & x > 0 \\ 0 \end{cases}$$

- This pdf corresponds to an exponential distribution with a mean of 2 years.
- To find the probability that the device's life falls between 2 and 3 years, we evaluate the integral of the pdf from 2 to 3:

$$P(2 \leq X \leq 3) = \frac{1}{2}\int_{2}^{3} e^{-x/2}dx$$

$$= -e^{-3/2} + e^{-1}$$

$$= -0.223 + 0.368$$

$$= 0.145$$

Cumulative Distribution Function

- The cumulative distribution function (CDF), represented by $f(x)$, shows the probability that a random variable X is less than or equal to a certain value x. Simply put, it tells us how likely it is for X to be at or below a specific point.
    - For a discrete random variable X, the CDF $f(x)$ is the sum of probabilities for all values of X that are less than or equal to x.

$$f(x) = \sum_{all\ x_i \leq x} p(x_i)$$

- For a continuous random variable X, the CDF $f(x)$ is found by integrating the probability density function $f(t)$ from negative infinity to x.

$$f(x) = \int_{-\infty}^{x} f(t)dt$$

- Properties of CDF
  1. The CDF is always increasing. This means that if you compare two values, say a and b, where a is less than b, then the CDF at a $f(a) < f(b)$.
  2. As x approaches positive infinity, the CDF approaches 1 $\lim_{x \to \infty} f(x) = 1.$

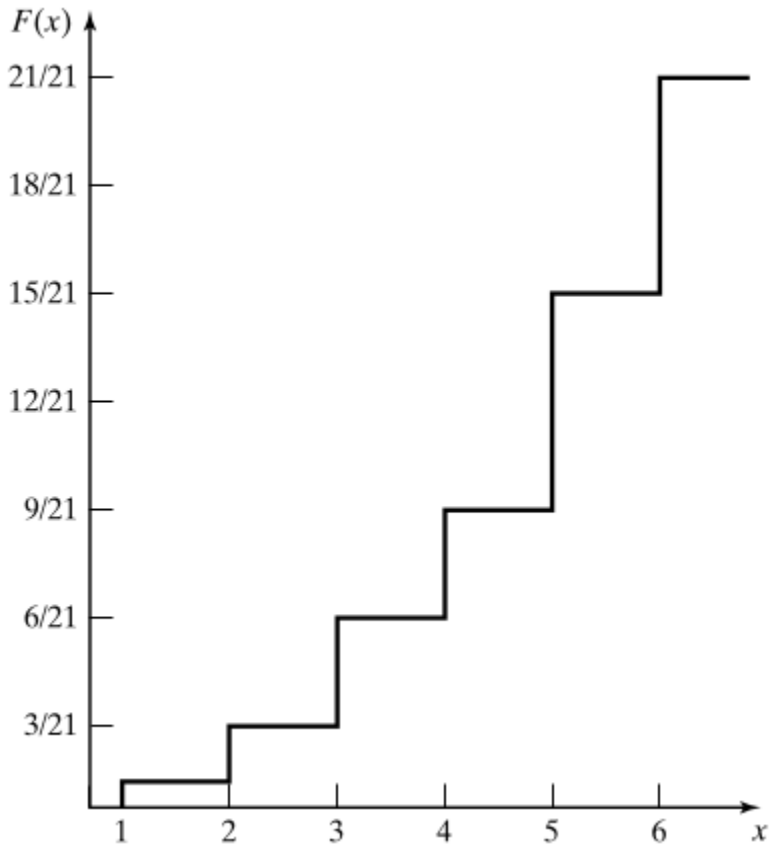
     This means that the probability of X being less than or equal to a very large number is close to 1.
  3. As x approaches negative infinity, the CDF approaches 0 $\lim_{x \to -\infty} f(x) = 0.$

     This indicates that the probability of X being less than or equal to a very small (or very large negative) number is close to 0.
- You can answer many probability questions about X using the CDF. For instance, if you want to find the probability that X falls between two values a and b, you subtract the CDF at a from the CDF at b.
- The cumulative distribution function (CDF) for the die-tossing experiment is given as follows:

x	$(-\infty, 1)$	$[1, 2)$	$[2, 3)$	$[3, 4)$	$[4, 5)$	$[5, 6)$	$[6, \infty)$
$f(x)$	0	1/21	3/21	6/21	10/21	15/21	1

- In this notation, [a, b) represents the interval where a is included, but b is not. For example, [1, 2) includes all values from 1 up to (but not including) 2.

- The CDF for this example is a step function, meaning it increases in steps rather than smoothly. At each value x_i, where x_i is one of the possible outcomes of the random variable X, the CDF increases by the probability $p(x_i)$ associated with that outcome.
- For instance, at $x = 3$, the probability $p(3)$ = 3/21, which is the size of the step or jump in the CDF. This jump signifies the increase in probability at the value 3.



- The cumulative distribution function (CDF) for the lifetime of the aircraft inspection device is given by:

$$f(x) = \frac{1}{2}\int_0^x e^{-t/2}dt = 1 - e^{-x/2}$$

- Using this CDF, we can find probabilities associated with the device's lifespan.
- The probability that the device will last for less than 2 years is calculated as:

$$P(0 \leq X \leq 2) = f(2) - f(0) = f(2) = 1 - e^{-1} = 0.632$$

- The probability that the device's lifespan is between 2 and 3 years is calculated as:

$$P(2 \leq X \leq 3) = f(3) - f(2) = (1 - e^{-3/2}) - (1 - e^{-1}) = 0.145$$

Expectation

- An essential concept in probability theory is the expectation of a random variable. If X is a random variable, its expected value, denoted by $E(X)$, is defined differently for discrete and continuous variables:
  - $E(X) = \sum_{all\ i} x_i . p(x_i)$ if X is discrete
  - $E(X) = \int_{-\infty}^{\infty} x . f(x) dx$ if X is continuous

- The expected value $E(X)$ is also known as the mean or the first moment of X. The quantity $E(X^n)$, where $n \geq 1$, is called the nth moment of X and is computed similarly.
  - $E(X^n) = \sum_{all\ i} x_i^n . p(x_i)$ if X is discrete
  - $E(X^n) = \int_{-\infty}^{\infty} x^n . f(x) dx$ if X is continuous

- The variance of a random variable X, denoted by $V(X)$ or $var(X)$ or σ^2, is defined as:

$$V(X) = E[(X - E[X])^2]$$

- A useful identity for computing $V(X)$ is

$$V(X) = E(X^2) - [E(X)]^2$$

- The mean $E(X)$ represents the central tendency of X, while the variance measures the expected squared difference between the variable and its mean.
- Therefore, the variance $V(X)$ quantifies the spread or variation of X around its mean.
- The standard deviation, σ, is defined as the square root of the variance. Both the mean, $E(X)$, and the standard deviation, $\sigma = \sqrt{V(X)}$, are expressed in the same units.
- The mean and variance of the die-tossing experiment from Example 2 are calculated as follows:
    - The mean $E(X)$

$$E(X) = \frac{1}{21} \times 1 + \frac{2}{21} \times 2 + \ldots + \frac{6}{21} \times 6 = \frac{91}{21} = 4.33$$

    - For variance, first we find $E(X^2)$

$$E(X^2) = \frac{1}{21} \times 1^2 + \frac{1}{21} \times 2^2 + \ldots + \frac{6}{21} \times 6^2 = 21$$

Therefore,

$$V(X) = 21 - (4.33)^2 = 21 - 18.7489 = 2.22$$

    - Standard deviation $\sqrt{2.22} = 1.49$

- The mean and variance of the device's life, are computed as follows:

$$E(X) = \frac{1}{2}\int_0^{\infty} x.\, e^{-x/2} dx$$

- To solve the integral, we can use integration by parts. Integration by parts is a technique that reverses the product rule for differentiation. It states:

$$\int u\, dv = uv - \int v\, du$$

We let $u = x,\ dv = \frac{1}{2}e^{-x/2}$,

Then we calculate $du = dx,\ v = -e^{-x/2}$

Now integration by parts

$$E(X) = \left[-xe^{-x/2}\right]_0^{\infty} - \int_0^{\infty}(-e^{-x/2})dx$$

The first term evaluates to $0 - \lim_{0 \to \infty}(-e^{-x/2}) = 0$, because $e^{-\infty} = 0$

We are now left with $E(X) = \int_0^{\infty} e^{-x/2} dx$